

What is AWS EBS Services

Amazon EBS is a persistent, high-performance block storage service for EC2. It behaves like a virtual hard drive, designed for workloads needing consistent I/O and low latency, such as:

Databases (MySQL, PostgreSQL, SQL Server, Oracle)

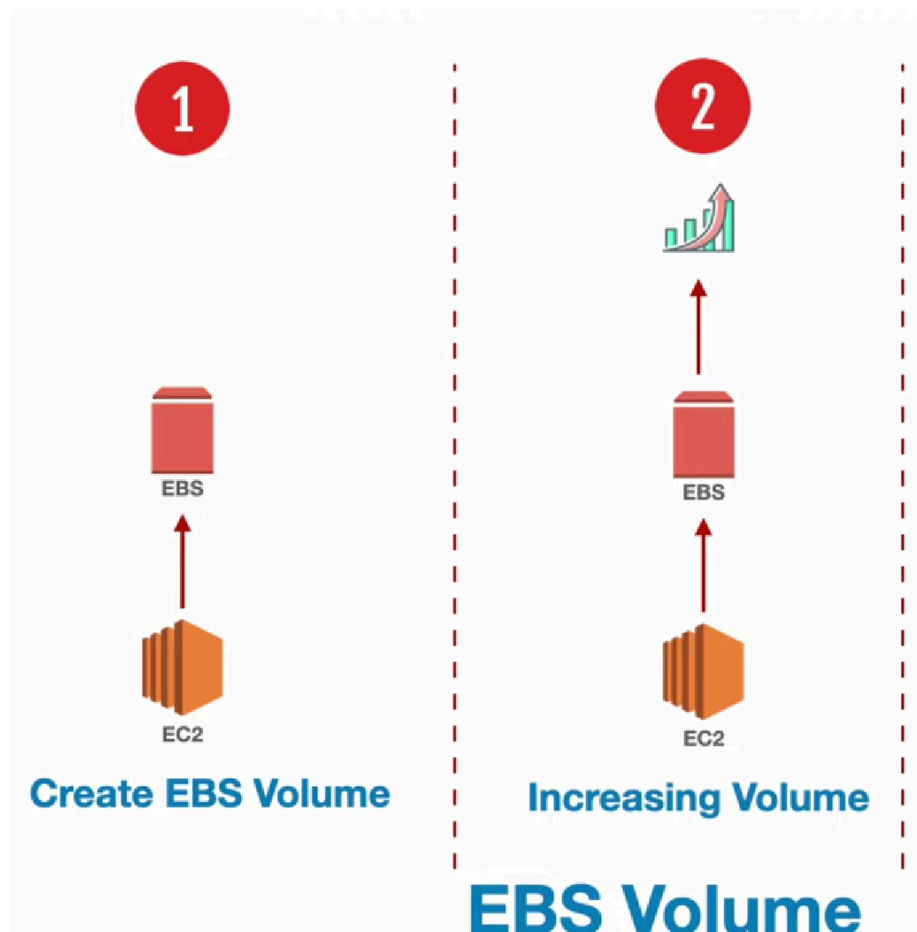
Kubernetes persistent volumes (EKS/ECS)

File systems and transactional workloads

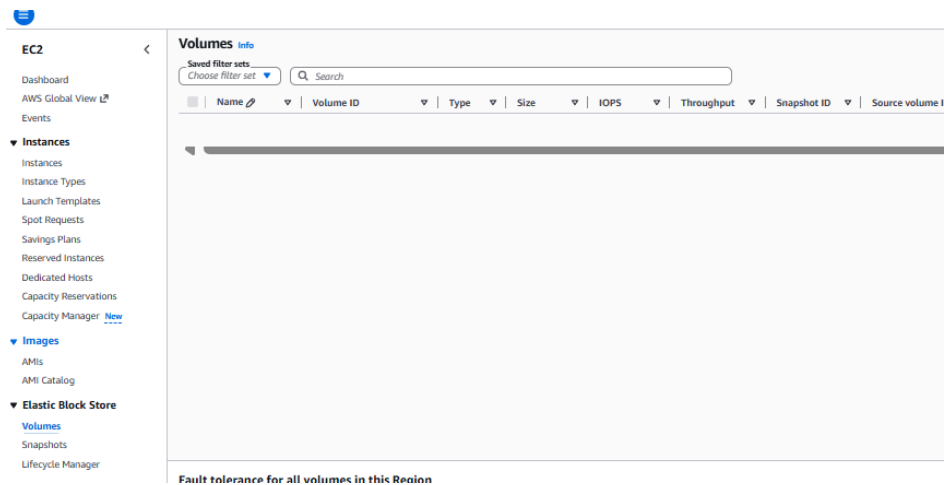
High-performance workloads requiring predictable IOPS.

ie.SSD,Provisioned IOPS SSD (IO1)Cold HDD (SC1), Throughput optimized HDD (ST),General Purpose SSD (GP3).

Amazon EBS is AZ-specific block storage that must be created and attached within the same Availability Zone as the EC2 instance.



Go to ec2 dashboard>scroll to volume>create



Create volume

Steps:-

Volume type:-SSD(TYPE:-gp1,2,3)

Size (GiB)=10gb /or as per requirement

IOPS=3000

Throughput (MiB/s)=125

Availability Zone=asia-mumbai-1a

.....create

EC2 > Volumes > Create volume

Create volume [info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [info](#)
General Purpose SSD (gp3)

Size (GiB) [info](#)
10
Min: 1 GiB, Max: 65536 GiB.

IOPS [info](#)
3000
Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [info](#)
125
Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone [info](#)
aps1-az1 (ap-south-1a)

Snapshot ID - optional [info](#)
Don't create volume from a snapshot

Encryption [info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

Volume type:-SSD(TYPE:-gp1,2,3)
Size (GiB)=10gb /or as per requirment
IOPS=3000
Throughput (MiB/s)=125
Availability Zone=asia-mumbai-1b

.....create

Volume settings

Volume type | [Info](#)

General Purpose SSD (gp3) ▼

Size (GiB) | [Info](#)

10

Min: 1 GiB, Max: 65536 GiB.

IOPS | [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) | [Info](#)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone | [Info](#)

aps1-az3 (ap-south-1b) ▼

Snapshot ID - optional | [Info](#)

Don't create volume from a snapshot ▼



Launch an ec2 instance
Add availability zone=ap-aisan-mumbai (as same as
create volume)

▼ Network settings [Info](#)

VPC - required | [Info](#)

vpc-060d64bd6f8be855
172.31.0.0/16

(default) ▼



Subnet | [Info](#)

No preference ▼



[Create new subnet](#)

Availability Zone | [Info](#)

ap-south-1a

aps1-az1 ▼



[Enable additional zones](#)

Auto-assign public IP | [Info](#)

Enable ▼

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Creating ec2 come with default 8 gb as root volume
(containe os,envirment,etc)
Add one more volume to different

▼ Configure storage [Info](#)

Advanced

1x

8

GiB

gp3

Root volume, 3000 IOPS, Not encrypted

1x

6

GiB

gp3

EBS volume, 3000 IOPS, Not encrypted

Remove

Add new volume

Launch an instance

linux-ecs-test

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

aws

Mac

ubuntu

Microsoft

Red Hat

SUSE

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-019715e0d74f695be (64-bit (x86)) / ami-0848881f2a3dcebd1 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

2 volume(s) - 14 GiB

Cancel

Launch instance

Preview code

Now connect the instances

EC2 Instances i-0178408b70e679d9d Connect to instance

Instance ID i-0178408b70e679d9d (linux-ecs-test) VPC ID vpc-060d64bd6f8be855 Security groups sg-0c3b894b136a4c771 (launch-wizard-3) IAM role ~

EC2 Instance Connect SSM Session Manager SSH client EC2 serial console

Instance ID i-0178408b70e679d9d (linux-ecs-test)

Connection type

Connect using a Public IP

Connect using a public IPv4 or IPv6 address

Connect using a Private IP

Connect using a private IP address and a VPC endpoint

Public IPv4 address

13.233.99.200

IPv6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

Q ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

Check the volume

Volume state:-root one in use and adding one available

Volumes (1/4) Info

Last updated 1 minute ago

Recycle Bin

Actions

Create volume

Saved filter sets

Choose filter set

Search

Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone
vol-0eef1a0b4f127be7b	gp3	10 GiB	3000	125	-	-	2026/01/25 22:13 GMT+5:...	aps1-az1 (ap-south-1a)
vol-06c9ecbc71664e029	gp3	8 GiB	3000	125	snap-05228cd...	-	2026/01/25 22:25 GMT+5:...	aps1-az1 (ap-south-1a)
vol-04dcd47b399fca1a3	gp3	6 GiB	3000	125	-	-	2026/01/25 22:25 GMT+5:...	aps1-az1 (ap-south-1a)
vol-0986e155e55d5b366	gp3	10 GiB	3000	125	-	-	2026/01/25 22:20 GMT+5:...	aps1-az3 (ap-south-...

Check the volume

Volume state:-root one in use and adding one available

```
ubuntu@ip-172-31-33-235:~$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                 7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1                 7:1    0   74M  1 loop /snap/core22/2163
loop2                 7:2    0 50.9M  1 loop /snap/snapd/25577
nvme1n1              259:0    0    6G   0 disk
nvme0n1              259:1    0    8G   0 disk
├─nvme0n1p1          259:2    0    7G   0 part /
├─nvme0n1p14         259:3    0    4M   0 part
├─nvme0n1p15         259:4    0 106M   0 part /boot/efi
└─nvme0n1p16         259:5    0 913M   0 part /boot
ubuntu@ip-172-31-33-235:~$
```

Attach the volume that we create

Ec2 dashboard>volume>select the volume
>action>attach volume

EC2 > Volumes > vol-09b27104200a13ca9 > Attach volume

Attach volume Info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-09b27104200a13ca9

Availability Zone

aps1-az1 (ap-south-1a)

Instance

i-0178408b70e679d9d (linux-ecs-test) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name

Select a device name

Recommended device names for Linux: /dev/sda1 for root volume, /dev/sd[f-p] for data volumes.

Cancel

Attach volume

After attach use same command lsblk

```
ubuntu@ip-172-31-47-2:~$ ls
ubuntu@ip-172-31-47-2:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0   27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1        7:1      0    74M 1 loop /snap/core22/2163
loop2        7:2      0   50.9M 1 loop /snap/snapd/25577
nvme1n1      259:0    0     6G 0 disk
nvme0n1      259:1    0     8G 0 disk
├─nvme0n1p1  259:2    0     7G 0 part /
├─nvme0n1p14 259:3    0     4M 0 part
├─nvme0n1p15 259:4    0   106M 0 part /boot/efi
└─nvme0n1p16 259:5    0   913M 0 part /boot
ubuntu@ip-172-31-47-2:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0      0   27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1        7:1      0    74M 1 loop /snap/core22/2163
loop2        7:2      0   50.9M 1 loop /snap/snapd/25577
nvme1n1      259:0    0     6G 0 disk
nvme0n1      259:1    0     8G 0 disk
├─nvme0n1p1  259:2    0     7G 0 part /
├─nvme0n1p14 259:3    0     4M 0 part
├─nvme0n1p15 259:4    0   106M 0 part /boot/efi
└─nvme0n1p16 259:5    0   913M 0 part /boot
nvme2n1      259:6    0    10G 0 disk
ubuntu@ip-172-31-47-2:~$
```

Now the disk is attach but we have to make it useable like disk partition

Common used `sudo fdisk -l`

```
Disk /dev/nvme0n1: 8 GiB, 8589934592 bytes, 16777216 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: gpt
Disk identifier: AB9BB1DA-4173-40B6-8D1C-300A500AC436

Device            Start      End  Sectors  Size Type
/dev/nvme0n1p1    2099200 16777182 14677983   7G Linux filesystem
/dev/nvme0n1p14    2048    10239    8192     4M BIOS boot
/dev/nvme0n1p15   10240   227327   217088  106M EFI System
/dev/nvme0n1p16   227328 2097152 1869825   913M Linux extended boot

Partition table entries are not in disk order.

Disk /dev/nvme1n1: 6 GiB, 6442450944 bytes, 12582912 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/nvme2n1: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
ubuntu@ip-172-31-38-138:~$
```

We can view our 10 gb disk volume is attach properly,important thing; we have t copy the path of the disk /dev/nvme2n1

```
ubuntu@ip-172-31-38-138:~$ sudo file -s /dev/nvme2n1
/dev/nvme2n1: data
ubuntu@ip-172-31-38-138:~$
```

Command use :- sudo file -s /dev/nvme2n1
Result =data (means, their is no file system.)

So,we have to give the file system command = (sudo mkfs -t xfs /dev/nvme2n1)

```
ubuntu@ip-172-31-38-138:~$ sudo mkfs -t xfs /dev/nvme2n1
meta-data=/dev/nvme2n1          isize=512    agcount=16, agsize=163840 blks
       =                       sectsz=512   attr=2,    projid32bit=1
       =                       crc=1        finobt=1, sparse=1, rmapbt=1
       =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
data      =                       bsize=4096  blocks=2621440, imaxpct=25
       =                       sunit=1      swidth=1 blks
naming    =version 2           bsize=4096  ascii-ci=0, ftype=1
log        =internal log      bsize=4096  blocks=16384, version=2
       =                       sectsz=512   sunit=1 blks, lazy-count=1
realtime  =none               extsz=4096  blocks=0, rtextents=0
ubuntu@ip-172-31-38-138:~$ sudo file -s /dev/nvme2n1
/dev/nvme2n1: SGI XFS filesystem data (blksz 4096, inosz 512, v2 dirs)
ubuntu@ip-172-31-38-138:~$
```

Now,we can see the the file system is given to our disk with same command.

Mount the disk to the file/directory.

Sudo mkdir /ebs1a

Sudo mount /dev/nvme2n1 /ebs1a

Ls -lart /ebs1a/

```

ubuntu@ip-172-31-38-138:~$ sudo mkdir /ebs1a
ubuntu@ip-172-31-38-138:~$ sudo mount /dev/nvme2n1 /ebs1a
ubuntu@ip-172-31-38-138:~$ ls -lart /ebs1a/
total 4
drwxr-xr-x  2 root root    6 Jan 26 14:41 .
drwxr-xr-x 23 root root 4096 Jan 26 14:47 ..
ubuntu@ip-172-31-38-138:~$ █

```

df -h (To view the space occupied by (xvf file system) on disk)

```

ubuntu@ip-172-31-38-138:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  1.8G  5.0G  27% /
tmpfs            458M    0  458M   0% /dev/shm
tmpfs            183M  900K  182M   1% /run
tmpfs            5.0M    0   5.0M   0% /run/lock
efivarfs         128K   3.9K  120K   4% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M   89M  730M  11% /boot
/dev/nvme0n1p15  105M   6.2M   99M   6% /boot/efi
tmpfs            92M   12K   92M   1% /run/user/1000
/dev/nvme2n1     10G  228M   9.8G   3% /ebs1a
ubuntu@ip-172-31-38-138:~$ █

```

2. To modify the volume

Important point we can increased the size of a volume but, cannot decrease the volume size or data lost will occur.

Volume >select the volume> Action>Modify

☰ [EC2](#) > [Volumes](#) > [vol-028de03a52fa70757](#) > **Modify volume**

Volume ID
 [vol-028de03a52fa70757 \(ebs-1a\)](#)

Volume type | [Info](#)

General Purpose SSD (gp3) ▼

Size (GiB) | [Info](#)

15

Min: 1 GiB, Max: 65536 GiB.

IOPS | [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) | [Info](#)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Sudo fdisk -l(to view the update of the disk partition)

```
sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes

Disk /dev/nvme2n1: 15 GiB, 16106127360 bytes, 31457280 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
ubuntu@ip-172-31-38-138:~$
```